

학부생을 위한 해석학 개론 (TP309-01)

학기	2026년 1학기
담당교수	안흥주
이수구분	교선평필수
학점	3.0
강의시간	화09:00-10:30(E7 - 223), 목09:00-10:30(E7 - 223)
강의실	E7 218 (교수실)
소속	기초학부

교과개요

This course provides a foundational understanding of mathematical analysis, focusing on rigorous reasoning and its applications to real-world problems in engineering and science. Students will explore the core concepts of limits, continuity, differentiation, and integration, emphasizing their practical relevance and theoretical underpinnings. The course bridges the gap between computational calculus and abstract mathematics, equipping students with the skills to analyze and model complex systems using precise mathematical methods. While focusing on understanding key analytical tools, the course also demonstrates their application to scientific and engineering problems, preparing students for advanced topics in applied mathematics and related disciplines.

교과목표

- Develop a rigorous understanding of limits, continuity, and differentiability.
 - Apply analytical techniques to solve problems in physics, engineering, and data modeling.
 - Strengthen logical reasoning and proof-writing skills.
 - Understand the convergence of sequences and series and their role in approximations.
 - Gain insight into real-world systems through integration and optimization applications.
- Details
- Understanding Basic Set Theory and Logic in Mathematics: Gain an understanding of fundamental concepts in set theory and logic used in mathematics
 - Real Numbers, Definition and Properties: Understand the definition and properties of real numbers, especially the concept of the least upper bound (supremum) and Learn how various theorems related to the convergence of sequences and series are closely connected to the least upper bound property
 - Limits and Continuity of Functions: Comprehend and apply the rigorous definitions of limits and continuity, including pointwise continuity and uniform continuity, and Prove related theorems, such as Intermediate and Extreme Value Theorem
 - Topological Properties in Euclidean Space: Understand various topological properties of Euclidean space and Recognize how this knowledge plays a crucial role in advanced mathematical courses such as topology
 - Differentiation of Single-Variable Functions: Understand and apply various properties related to the differentiation of single-variable functions
 - Definition and Extension of Integration
 - 1) Define differentiation and integration based on the rigorous definition of limit
 - 2) Expand the concept of integration to the Riemann-Stieltjes integral
 - 3) Prove related theorems such as L'Hospital's Rule, the Fundamental Theorem of Calculus, and Darboux's Theorem.
 - 4) Extend the concepts of differentiation and integration to general multivariable functions.
 - Sequences of Function
 - 1) Understand and apply fundamental concepts of sequences of functions

- 2) Grasp the notions of pointwise convergence and uniform convergence
- 3) Identify the conditions under which operators such as differentiation, integration, summation, and limits can be interchanged.

주차별 강의계획

- 1주차: See the attached file
- 2주차:
- 3주차:
- 4주차:
- 5주차:
- 6주차:
- 7주차:
- 8주차:
- 9주차:
- 10주차:
- 11주차:
- 12주차:
- 13주차:
- 14주차:
- 15주차:
- 16주차:

교과정책(방침)

- 1 hour's informal practice sessions (almost) every week.
- English lecture in principle
- Up to 5 unexcused absences allowed. More than 6 unexcused absences (20%) may result in a lower grade than the original: Arriving more than 15 minutes after the start of class will be considered tardy. Three instances of tardiness will be counted as one absence. A maximum of five absences is allowed, including for personal events and natural disasters. If a student is absent without a valid reason more than five times, a lower grade than originally assigned may be given. Attendance will be combined for both lectures and practice sessions.
- Absolute evaluation in the class (절대평가)
- Failure to attend midterm and final exams will result in an F grade regardless of score (there is no make-up exam)
- Anything below 20 points out of 100 (20 is not an absolute score but with some variation depending on the tests) is an F.
- In the event of an unavoidable withdrawal by the student, credit may reasonably be granted, considering the points earned by the student, as evidenced by clear documentation
- For homework, after completing each chapter, list the following: (1) keywords, (2) at least one concept and one theorem that you consider most important, summarized, (3) a one-sentence summary of the entire chapter, and (4) solve and submit at least one related problem. Use the provided template, but the length should not exceed 2 pages of A4 paper (anything beyond 2 pages will not be graded).

-매주 비공식 연습시간 1시간 운영예정

-강의는 원칙적으로 영어로 진행

-수업 시작 15분 이후 입실할 경우 지각으로 간주. 지각 3번은 결석 1번으로 간주. 경조사 및 천재지변을 포함하여 5번까지 결석 허용. 5회 이상 정당한 사유 없이 무단 결석한 경우, 본래보다 낮은 학점이 부여될 수 있다. 강 의와 연습 출석을 합산

-중간, 기말 고사 중 하나를 치르지 않으면 점수와 관계없이 F 처리됨

-매우 중요: 숙제는 매 장(chapter)이 끝나면 그 장의 (1) key words 를 열거하고, (2) 가장 중요하다고 생각되는 개념과 정리를 각각 1개 이상 선택하여 정리하고, (3) 각 장의 전체 내용을 한 문장으로 요약 하고, (4) 관련 문제를 1개 이상 풀어서 제출. 주어진 양식을 이용하되 분량은 A4 용지 2pages 이하로(2page 초과분은 채점하지 않음)함

-숙제 내용을 적을 때 다른 사람, 책, 논문 등 아이디어를 빌린 곳이 있으면 반드시 관련 내용을 명시할 것.

- 개념과 정리의 내용을 묻는 퀴즈가 있을 수 있음. 숙제는 정해진 기한내에 제출해야 하며 늦게 제출하는 숙제는 감점하며, 요구한 내용을 채우지 내용이 우수할수록(개수와는 관계없음) 높은 점수를 받을 수 있음.
- 원칙적으로 모든 시험은(퀴즈 포함) 재시험 없음.